



GC-NET for classification of glaucoma in the retinal fundus image

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Abstract

Glaucoma is the second-most dominant cause for irreversible blindness, resulting in damage to the optic nerve. Ophthalmologist diagnoses this disease using a retinal examination of the dilated pupil. Since the diagnosis is a manual and laborious procedure, an automated approach for faster diagnosis is desirable. Convolutional neural networks (CNN) could allow automation of the diagnosis procedure due to their self-learning capabilities. This paper presents a deep learning-based glaucoma classification network (GC-NET) for classifying a retinal image as glaucomatous or non-glaucomatous. The proposed GC-NET has been tested on RIM-One and Drishti datasets. Our experimental results showed that GC-NET achieves accuracy of 97.51%, sensitivity of 98.78% and specificity of 96.20% with 0.81 true positive (T_p), 0.03 false positive (F_p), 0.76 true negative (T_n) and 0.01 false negative (F_n) which outperforms state of the art. Thus, the proposed approach could be very useful for initial screening of glaucoma patients.

Keywords Glaucoma · Deep learning · Computer-aided diagnosis · Retinal fundus image

1 Introduction

Glaucoma well-known as “silent theft of sight” was probably recognized as a disease in the initial years of 17th century, when the greek term *glaukEoma* meaning obscurity of the lens or cataract signifying lack of understanding of the disease came into existence. It begins due to an increase in the pressure of an eye, known as intraocular pressure leading to damage of the optic nerve. While optic nerve carries visual information via retina to the brain resulting in visualization of the outside world [1]. It is the common

cause of blindness, which has affected more than 60 million people till date across the globe and may rise to 80 million by 2020. According to one of the studies, it is estimated to influence 12 million people in India, which is around one-fifth of the total Indian population. Its common symptoms include sudden visual disturbance, severe eye pain, blurred vision, red eyes and haloes around lights. This disease is normally found to affect people with age more than 40 years [2]. Ophthalmologists use different comprehensive examinations such as ophthalmoscopy, tonometry, perimetry, gonioscopy and pachymetry to diagnose glaucoma. Tonometry is the measure of internal eye pressure, whereas ophthalmoscopy is the examination of the colour and shape of the optic nerve. Perimetry is the visual field analysis, pachymetry is the measure of corneal thickness, and gonioscopy is the measure of the angle between the iris and cornea [3]. All these approaches are manual, consume time and may lead to biased decisions by different experts. Thus, there is a need for a computer-aided diagnosis (CAD) system which can act as a second opinion for doctors. CAD systems for glaucoma can use retinal fundus imaging as an input to extract different types of features and classify the retinal image as “abnormal” or “normal” [4]. Retinal fundus image is acquired using a fundus camera which consists of a microscope attached with flash-enabled camera to capture the image of the interior surface/backside of an eye. The image comprises of the

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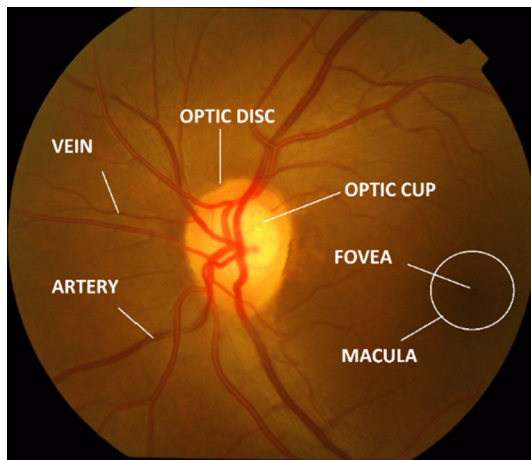


Fig. 1 Retinal fundus image

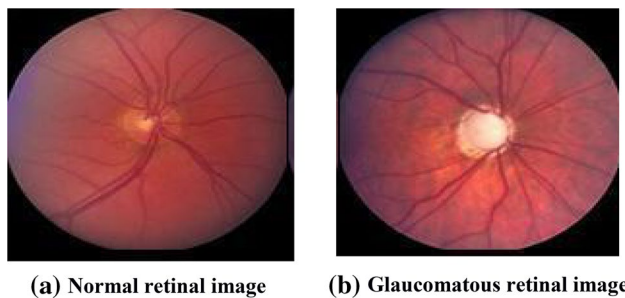


Fig. 2 Normal and abnormal retina

optic cup, optic disc and retinal vessels (arteries, veins) as shown in Fig. 1.

Here, the optic disc is the centric bright portion of retina where various optic nerves come together, and the optic cup is the depression of changeable size present on the optic disc. Glaucoma can be diagnosed by analysing the progression of the optic cup, which covers the optic disc completely in cases of permanent blindness. Whereas retinal vessels, including veins and arteries, carry blood to different parts of an eye. Also, the fovea is the small concave-shaped area present in the centre of the retina without vessels. Lastly, the macula is the dark pigmented area, yellowish in colour present on the retina with an approximate diameter of 5.5 mm [5]. Figure 2 shows the normal and abnormal retinal image with the varying optic cup.

Medical imaging has been employing computers for diagnosis of abnormality using imaging modality which is a non-invasive approach. Similarly, an effective way to reduce the further progression of glaucoma resulting in blindness can be taken into consideration, which necessitates diagnosis at an early stage using computer-aided diagnosis system. Typically, the diagnosis of glaucoma

performed by ophthalmologists includes contouring of cup and disc in the retinal imaging manually to calculate the disc damage likelihood scale (DDLS), inferior superior nasal temporal (ISNT) order and cup-to-disc ratio (CDR). Followed by these few other physical examinations such as tonometry, pachymetry, gonioscopy etc. are performed to confirm the disease with its stage to prescribe treatments to patients accordingly. But all these approaches are manual, tiresome and prone to subjective errors; thus, a computer-aided diagnosis system based on deep learning can be used to overcome the drawbacks of traditional approaches. The development of CAD systems for glaucoma can assist ophthalmologists by acting as a second opinion and save their precious time. It can reduce the workload of clinicians, minimize the risk of misclassification and improve the inter/intra-observability.

This work presents an automated approach to diagnose glaucoma using convolutional neural network-based deep learning, as this does not require extraction of any features. A basic convolutional neural network (CNN) consists of single or multiple convolution layers followed by fully connected layers and is one of the most efficient ways of fully utilizing the 2D structure of the input such as an image or a signal. It includes multiple convolutional layers, pooling layers, normalization layers, upsampling and downsampling layers, and so on, to bring out the best results. Multiple steps are performed for the network to work efficiently and for it to produce a beneficial result [6].

Basic steps followed by CNN are:

- (i) **Padding:** Padding prevents the size of the image from decreasing and thereby retaining the actual size of the image. It prevents the image from disappearing and allows the input size and output size to be the same using Eq. (1)

$$\text{pad} = (f - 1)/2 \quad (1)$$

Here ' f ' represents the size of the filter. It is used to perform the convolution of filter ' f ' having size ' $n \times n$ ' with the image portion of size ' $n \times n$ '.

- (ii) **CNN Output:** The output of a convolutional neural network is determined using Eq. (2)

$$\text{Output} = ((\text{Input} + 2 \times \text{pad} - f)/s) + 1 \quad (2)$$

where 'Input' signifies input dimension, 'pad' is the padding size, ' f ' is the filter dimension and ' s ' is the size of stride.

- (iii) **CNN layer output:** Finally, the output of a convolution layer is computed using Eqs. (3) and (4)

$$z^n = W \cdot (x^{n-1}) + b^n \quad (3)$$

$$x^n = a^n \cdot z^n \quad (4)$$

where ‘ W ’ represents weights of the convolution layer, x^{n-1} is an intermediate value obtained from the previous layer, ‘ b ’ is the bias, ‘ x^n ’ signifies output of the convolution layer and $a^n[n]$ is the activation function.

Rest of the paper is further categorized into Sect. 2 with related work followed by Sect. 3 with deep learning-based classifiers. Section 4 presents the methodology of the proposed GC-NET, whereas Sect. 5 presents experimental results and discussion pursued by Conclusion in Sect. 6.

2 Related work

2.1 Machine learning approaches

Machine learning is an approach that allows the prediction of outcomes based on the training provided with features as input without explicit instructions [7]. Some of the classification approaches using machine learning are categorized into three broad categories based on the features extracted such as non-structural features, wavelet features and structural features:

2.1.1 Non-structural features

These are the features which emphasize on the internal features of the object, i.e. higher-order spectral (HOS), textural and statistical etc.

Some of the approaches which extracted these features are:

Acharya et al. extracted HOS and textural features from an input image and performed glaucoma diagnosis using sequential optimization (SO), support vector machine (SVM), random forest (RF) and Naive bayes (NB). Based on the experimental analysis, RF classifier was found to outperform SVM, SO and NB with an accuracy of 91% [8].

Later, Noronha et al. extracted HOS features for classification of the retinal image as abnormal and normal using SVM/NB classifier and achieved an accuracy of 92.65% [9].

Further, Acharya et al. gave decision-based system for glaucoma with a machine learning approach. The statistical features such as variance, mean, kurtosis, skewness etc. were extracted. Finally, the classification was performed using SVM and NB, where the performance of SVM was found to be 93.13% accurate [10].

Also, Haleem et al. gave an automated approach for classification of the retinal image as ‘abnormal’ and ‘normal’ using an SVM classifier. The input retinal image was pre-processed using morphological closing, and the weighted

feature maps were applied to perform localization of optic disc. Further, statistical shape modelling was used for segmentation of ISNT regions, and other features such as Gaussian, texture, gabor, wavelets were extracted for classification. Thereafter, the features extracted were normalized using z-score normalization, and the classification was done using SVM classifier with an accuracy of 94.4% [11].

Claro et al. presented classifiers, namely maximum linear progression (MLP), random committee, random forest, and analysed that MLP outperforms others with an accuracy of 93.03%. Segmentation of disc was performed by thresholding, and grey-level run length matrix (GLCM) features were extracted for classification of abnormal and normal cases. Finally, features extracted were fed into different classifiers for testing and the performance of each was analysed using different performance metrics [12].

Later in De Sousa et al. applied equalization of the histogram to enhance the contrast of an image and used local binary patterns (LBP) to get a modified image representation. Further, they extracted geostatistical features, namely semivariogram, covariogram and correlogram to classify retinal images as abnormal and normal. The features extracted were then normalized in the range of -1 to 1, and the significant features were extracted using a micro-genetic algorithm. Finally, the classification was carried out using SVM with 91.2% accuracy [13].

Recently, Selvathi et al. performed enhancement to improve the quality of the image and applied wavelet transform to extract features in four directions (0°, 45°, 90°, 135°). Also, statistical features such as energy, standard deviation, mean and variance were extracted, and the classification was performed using artificial neural network (ANN). Based on the experimental evaluation, the accuracy of ANN was observed to be 95.8% [14].

2.1.2 Wavelet features [15]

Wavelet transformation is an approach that analyses data according to resolution or scale in different windows. “Wavelets” refers to a set of orthonormal basis functions produced by translation and dilation of mother wavelet Ψ and scaling function ϕ . The wavelet or orthonormal basis function is given as:

$$\Psi_{(j,k)}(x) = 2^{j/2} \Psi(2^j x - k) \quad (5)$$

and scaling function is defined using Eq. (6)

$$\phi_{(j,k)}(x) = 2^{j/2} \phi(2^j x - k) \quad (6)$$

Here, ‘ j ’ is the scaling and ‘ k ’ is the dilating integer. Further, in terms of wavelet coefficients, Eq. (5) is also represented as:

$$\Psi(x) = \sum_k^{N-1} g_k \sqrt{2\phi(2x-k)} \quad (7)$$

Here, $g_0, g_1, \dots$ are wavelet coefficients of high pass filter. Also, the scaling coefficients are given as:

$$\phi(x) = \sum_k^{N-1} h_k \sqrt{2\phi(2x-k)} \quad (8)$$

where $h_0, h_1, \dots$ are scaling coefficients of low pass filters. The scaling and wavelet coefficients related by the relation of quadrature mirror is given as:

$$g_n = (-1)^n h_{1-n+N} \quad (9)$$

The term ' N ' here is the number of moments vanishing in Eq. (9).

Some of the approaches which extracted wavelet features for classification are as follows:

Dua et al. extracted wavelet-based features from input retinal image for classification of normal and abnormal cases of glaucoma. Based on the rankings, significant features were classified using SVM, SO, RF and NB. Finally, the performance of classifiers was analysed, and it was observed that SVM outperform others with an accuracy of 93% [16].

Also, Mookiah et al., extracted HOS and wavelet-based features for classification of glaucoma using SVM and its variants followed by the formulation of glaucoma risk index (GRI). The SVM with kernel function of second order was found to outperform others with 95% accuracy [17].

Singh et al. extracted centre of the optic disc using kaiser-hat window and applied bit plane slicing for optic disc segmentation. The vessels present in segmented optic disc were removed by replacement of median, and the wavelet features were extracted for diagnosis of glaucoma. Further, the significant features were selected using principal component analysis (PCA) and evolutionary attribute selection, and the classification was performed using random forest, NB, K nearest neighbour (k -NN), ANN and SVM. Based on the experimental analysis, the performance of PCA with SVM was found to outperform others with an accuracy of 94.7% [18].

Koh et al. suggested random forest for classification of input retinal images as abnormal and normal. They started with pre-processing of the retinal image by contrast limited adaptive histogram equalization (CLAHE) and applied continuous wavelet transform (CWT) to delineate wavelet features. Entropy features were then extracted, and particle swarm optimization (PSO) was applied for feature selection. Finally, the testing was performed, and 92.4% accuracy was achieved with RF classifier [19].

2.1.3 Structural features

These are the features which emphasize on the external features of the ROI detected, i.e. area, volume, length, breadth, perimeter, etc. Structural features employed for glaucoma diagnosis are CDR, DDLS and ISNT. Some of the approaches which extracted structural features are:

Issac et al. suggested classification of retinal images as normal and abnormal using SVM and ANN classifier. Further, the structural features like neuroretinal rim (NRR) area, CDR, ISNT vessels were extracted for classification using SVM and ANN. Based on the experimental analysis, SVM classifier was found to outperform ANN with 94.11% accuracy [20].

After that, Salam et al. used both structural and non-structural features to create a hybrid set of features for the diagnosis of glaucoma. The pre-processing was performed using contrast stretching, negation, morphological opening, region growing, canny edge detector and ellipse fitting. The structural features such as CDR, NRR area and ISNT were extracted from segmented regions, and the non-structural features such as texture, wavelets, local binary patterns were extracted from the original input image. Finally, the SVM was applied to get normal and abnormal cases, with 92% accuracy [21].

2.2 Deep learning-based approaches

With the recent advances in the neural network, deep learning-based approaches have profoundly increased due to self-learning skill and less user input, ability to work on unstructured data and delivering efficient result [22]. Some of the deep learning approaches used for diagnosis of glaucoma are as follows:

Benzebouchi et al., presented deep learning-based CNN for the diagnosis of glaucoma. The approach used two sets of CNNs, the first one with the RGB colour retinal image and the second one with the greyscale input retinal image binarized using otsu thresholding. Based on the evaluation of the CNN1 with 25 epochs and 10 iterations, accuracy was observed to be 95.6%, sensitivity to be 98.5% and specificity to be 93.3%. Whereas for the CNN2 with 25 epochs and 5 iterations, accuracy was observed to be 96.95%, sensitivity to be 96.5% and specificity to be 97.3% [23]. Thereafter in Bander et al. used CNN-based deep learning for extraction of features to classify the retinal image as 'abnormal' or 'normal' using an SVM classifier. The pre-trained Alexnet CNN model with 23 layers of convolution, max-pooling, fully connected, softmax and output was used for training of feature extraction. The weights learned from Alexnet were transferred as the weights of the network. Based on the experimental evaluation, accuracy is observed to be 88.2%, sensitivity to be 85% and specificity to be 90.8%

[24]. Recently, Chai et al. designed multibranch neural network (MB-NN) comprising of faster RCNN, fully convolutional neural network (FCN) and CNN. Each CNN model comprised of 5 layers with rectified linear units (ReLU) as an activation function. The designed model gave 91.5% accuracy, 92.3% sensitivity and 90.9% specificity [25]. Also, Naoto et al. also gave CNN-based approach for diagnosis of glaucoma using a deep residual learning algorithm. The comparison of the proposed approach was performed with SVM and random forest and based on the experimental analysis, residual network (ResNet) was found to outperform with area under receiving operator characteristics (AUROC) between 96.5% and 97% [26]. Raghavendra et al. applied eighteen-layer CNN to extract deep features through retinal fundus image for diagnosing glaucoma. The CNN used comprises of two parts, first is the convolutional and max-pooling layer with the input of each layer as an output of the previous layer. While the second part includes a fully connected layer with softmax activation function for classification of extracted features. Further, the testing was performed on 30% remaining data excluded from testing, where the values of accuracy, specificity and sensitivity for CNN were observed to be 98.13%, 98% and 98.30% [27]. Norouzfard et al. presented transfer-based deep learning models for classification of retinal images. The approach initiated with cropping of input retinal image followed by the use of ResNet-V2 and VGG19 models. ResNet-V2 comprises of layers such as output, input, convolution, pooling, concatenate, residual, dropout and other fully connected layers. Based on the experimental evaluations, with training on 70% of data and testing on 30% of data, the VGG19 model gave an accuracy of 92.3% on 30th epoch [28]. Similarly, Lima et al. performed diagnosis of glaucoma using CNN and applied Genetic Algorithm (GA) for optimization. The input retinal image was initially pre-processed using cropping to resize the image into 96×96 pixels. This image was then fed into 25 layers CNN model comprising of batch normalization with size 16 and 200 epochs, convolution, ReLU activation and dropout. Model of CNN was then optimized using neuroevolution of augmenting technologies (NEAT) which is a type of genetic algorithm. The approach achieved an accuracy of 91% on 30% of test images after training, with 70% of images comprising both normal and abnormal cases [29]. Further, Kim et al. presented a computer-aided diagnostic of glaucoma using deep learning. The approach used CNN and gradient weighted activation mapping (Grad-CAM) for localization and classification of retinal images. The VGG16 model was modified for training by changing the end layer of the basic model into a full convolution layer, where the weights were fine-tuned using a batch of size 32 at a learning rate of 0.0001. The training was performed on 50% of imaging dataset in VGG16 model over ImageNet with 80 epochs, and testing on rest 50% images

accounting in 91% accuracy [30]. Recently, Diaz-Pinto et al. presented an analysis of different CNN models such as VGG16, InceptionV3, VGG19, ResNet50 and Xception. Based on the experimental analysis, Xception was found to outperform other CNN models with an average AUROC of 0.96, a specificity of 0.85 and sensitivity of 0.93 [31]. Also, Gomez-Valverde et al. exploited different types of CNN for classification of glaucoma. The classification was performed using CNN with convolution, fully connected, pooling and softmax with batch normalized after each convolutional layer. Further, the testing was performed on VGG19, GoogLeNet, ResNet, DenseNet, and based on the experimentation VGG19 was found to outperform other networks with AUROC of 0.94, specificity of 89.01% and sensitivity of 87.01% [32].

As per the literature, it can be analysed that different features have been used from time to time for classification of glaucoma using varied classifiers and retinal imaging datasets. Thus, it becomes difficult to analyse the significant features and classifiers for the diagnosis of glaucoma. Moreover, machine learning-based approaches require features as an input to classifier for optimal training. Further, the selection of optimal features plays a significant role in classification. Thus, deep learning-based approaches may overcome these drawbacks as they automatically extract features and perform self-learning. Currently, the use of deep learning-based approaches is been employed for classification of glaucoma. But there is a need of an optimal approach with improved accuracy. Thus, a deep learning-based approach has been presented in this work to perform optimal diagnosis.

3 Deep learning-based classifiers

This section describes the implementation of various state of the art approaches used for comparison with the proposed approach.

3.1 InceptionV3

Inception is a commonly used model in the field of image recognition using convolution network. It is a 48-layer deep architecture having convolutional layers of filter size 3×3, maxpool layers of filter size 3×3 which propagate with a stride 2, batch normalization layers and inception layers considered as the main attribute of this architecture. Inception layer is a combination of 1x1, 3x3 and 5x5 convolutional layers with their output clubbed together to act as the input of the next layer. The tail of the architecture has a dropout layer and a dense layer having softmax as the activation function, which gives us the final output. It uses two 3x3 convolutional layers instead of a 5x5 convolutional layer

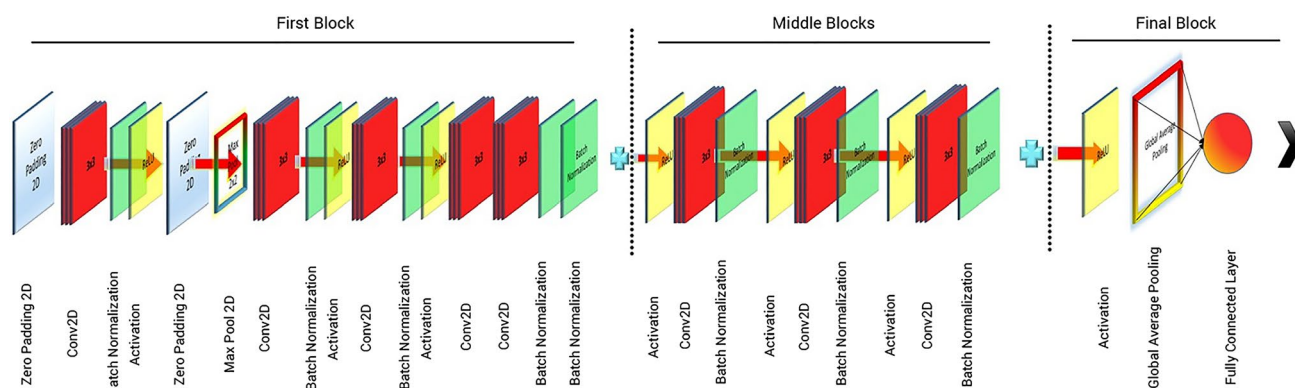


Fig. 5 ResNet 50 architecture

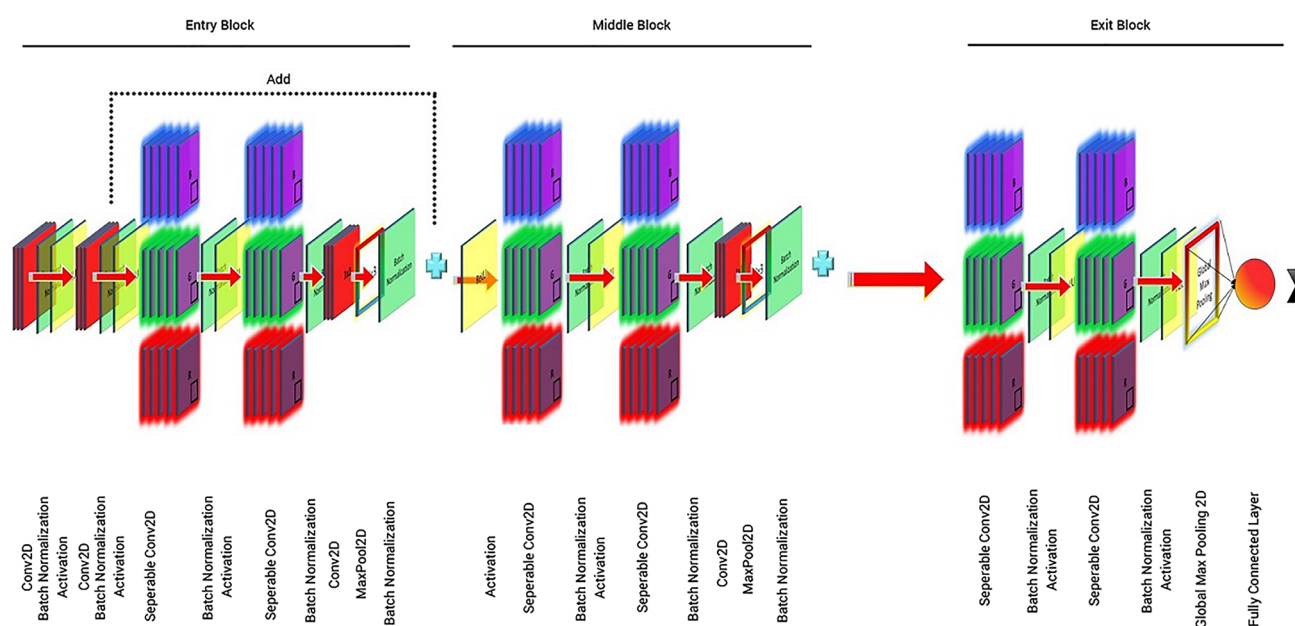


Fig. 6 Xception architecture

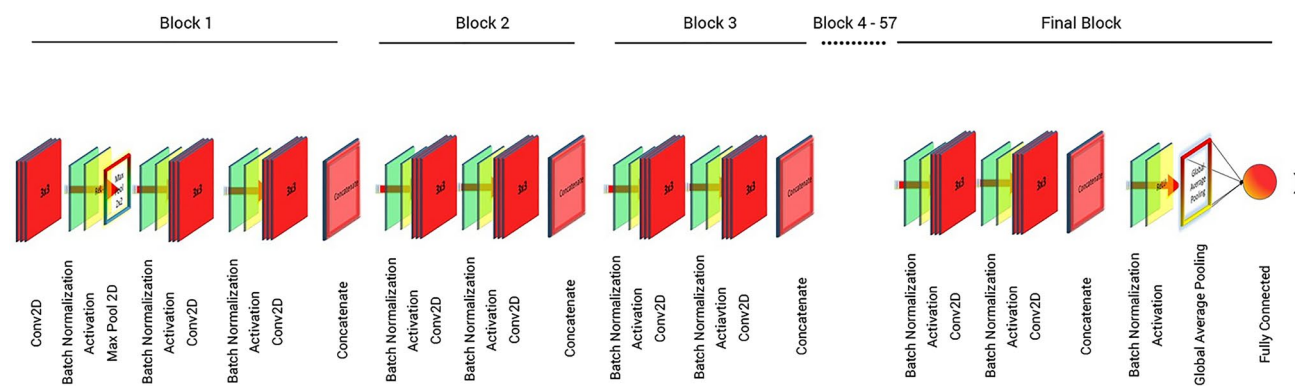


Fig. 7 DenseNet121 architecture

The block ends with a concatenation layer which concatenates the output of preceding layers. The tail of architecture consists of a dropout layer having a dropout rate of 0.2, followed by a dense layer of 1 unit and sigmoid activation [36]. The diagrammatic representation of DenseNet121 with blocks and layers is shown in Fig. 7.

3.5 Vgg16

VGG16 is a 16 layers deep CNN architecture having a stack of convolution layers with a reduced receptive field of filter size 3x3, and the input to the model is a 224×224 input RGB image. The stride of convolution is fixed as 1 pixel, along with spatial padding to preserve the resolution. The width of convolution layers starts with 64 filters and increases with a factor of two after every Max Pool layer, the last layer having a width of 512. Pooling is carried out by five Max Pool layers over a pool size of 2x2, with a stride 2. Non-linearity

is introduced by ReLU activation function. The architecture ends with a dropout layer having a dropout rate of 0.2. The tail of the model consists of three fully connected layers. The last fully connected layer consists of 1 unit and sigmoid activation [37]. The diagrammatic representation of VGG16 with blocks and layers is shown in Fig. 8.

4 Methodology

The proposed approach initiated with pre-processing of input retinal image by cropping, augmentation and denoising. Cropping was applied to reduce the size of the image and the computation time. The augmentation was performed by rotating, shifting and flipping to increase the number of images in the dataset. Denoising was performed to remove the additive and multiplicative noise present in retinal fundus images. Further, the pre-processed image was given as input to the proposed

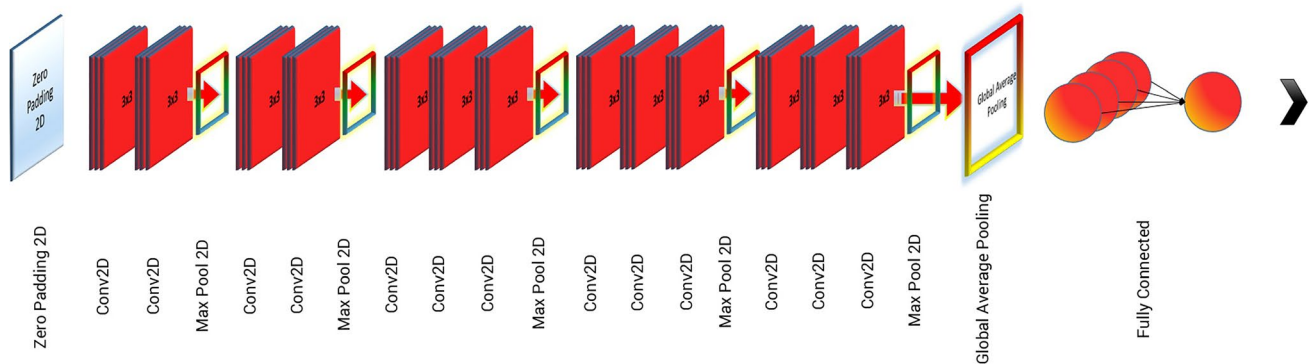


Fig. 8 VGG16 architecture

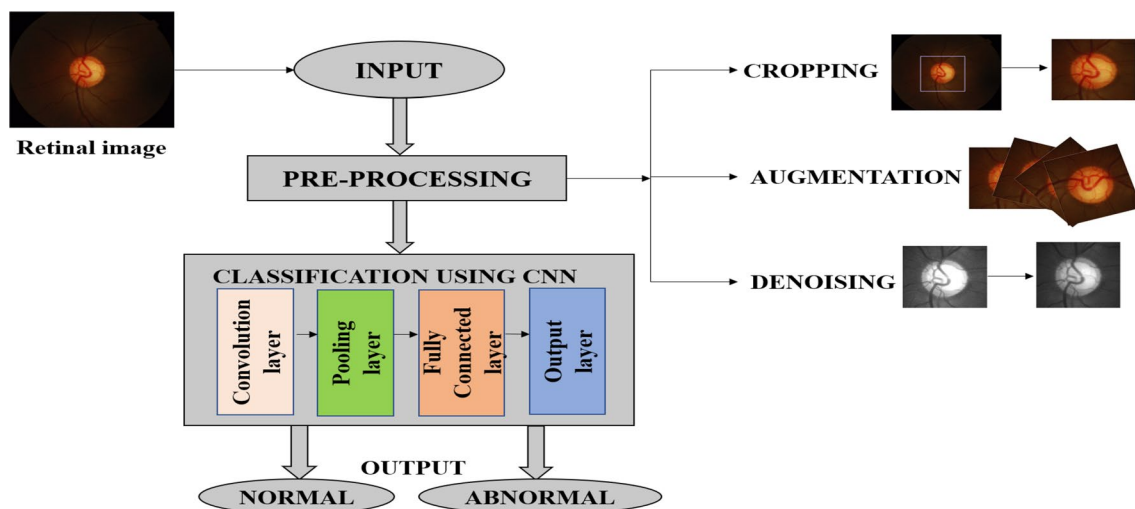


Fig. 9 Basic methodology

CNN model with 76 layers and trained by passing through different layers of convolution. Finally, the testing was performed, and predicted results were validated with ground truth labels, i.e. ‘normal’ and ‘abnormal’. Figure 9 presents the steps performed by the proposed GC-NET to classify the glaucomatous and normal image.

The detailed explanation of procedural steps are as follows:

4.1 Pre-processing [38]

The pre-processing was done to remove irrelevant details from the input data. Diagnosis of glaucoma using retinal imaging involves an analysis of pale yellowish region called the optic disc. The acquired image comprises of extra irrelevant details leading to an increase in the size of an image, as a result of which computation time is increased. Also, the presence of irrelevant details results in the reduction of accuracy, due to false regions and in case of CNN training requires a large number of input data for optimal performance. Thus, the pre-processing of the retinal image involves cropping the image to remove irrelevant details, augmentation to increase the number of images in the dataset for training convolution network. Further, the denoising is performed to remove the additive and multiplicative noise present in the retinal fundus images.

4.1.1 Cropping

The original fundus image of dimensions 2896×1944 pixels from the dataset is dimensionally quite large to be given directly as an input to the neural network. Therefore, in order to reduce the dimensions, the cropping of the targeted portion, involving the optic disc, to 512×512 pixels is done manually. Cropping does result in data loss, but since our primary focus is the optic disc and optic cup region, thus, this loss becomes almost irrelevant. Also, it leads to a reduction in computation time, brings about a major increase in the accuracy of the classification model and uniformity to the input image for further processing. Figure 10 shows the original and the input cropped image.

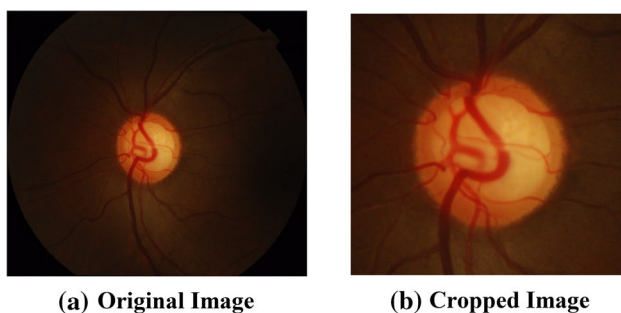


Fig. 10 Cropping of region of interest

4.1.2 Data augmentation

Data augmentation is an approach to increase the number of samples in the dataset to avoid overfitting. Deep learning-based approaches require training at large scale for correct prediction. It increases the number of images for training, as a result of which “better is the training improved is the accuracy of testing”. Thus, augmentation plays a significant role in the case of a smaller number of samples in the dataset. For instance, Zhang et al. used scaling, rotation, noise injection, translation and gamma correction for augmenting the images in MS brain dataset with scaling values from 0.7 to 1.3 [39]. Also, Zhang et al. employed rotation, gamma correction and noise addition for augmenting the dataset. Rotation from angle -15° to 15° at the step of 5° , gamma correction with $r = 0.7$ to 1.3 at the step of 0.1 and Gaussian noise with a variance of 0.01 was employed to increase the number of samples in the dataset [40].

But in this case, the original dataset consisted of only 267 images which could be used to train the neural network. These number of images were not enough to train the model and to be applied on a neural network. So, in order to tackle this issue, techniques like rotation, flipping and zooming were used for data augmentation. Firstly, the images were vertically flipped, which doubled the size of the dataset. Thereafter, the images were given a horizontal flip, which further increased the number of images in the dataset. Thereafter, the images were given a θ angle rotation. The angle of rotation θ was incremented from 0° to 180° with an increase in 20° at every step. This led to a significant increase in the size of data because of which the model’s training was done quite efficiently. The augmentation helps to improve the performance of the model, as it increases the number of images for training and better is the training of the model, better is the testing accuracy. Figure 11 shows the augmentation of images performed to increase the number of input images for the training of proposed GC-NET.

4.1.3 Denoising

Denoising is an approach to remove noise present in the images for better diagnosis of abnormality. Presence of noise degrades the quality of image, thereby suppressing the necessary details which may cause inaccurate diagnosis. As per the recent study performed by Elseid et al. [41], retinal fundus images are found to be degraded with Gaussian noise and salt and pepper noise. Further based on the analysis performed by them, the adaptive median filter is found to remove these noises and increase quality of image, thereby making it suitable for diagnosis. Thus, this study used the adaptive median filter to remove the Gaussian, salt and pepper noises present in the retinal fundus images.

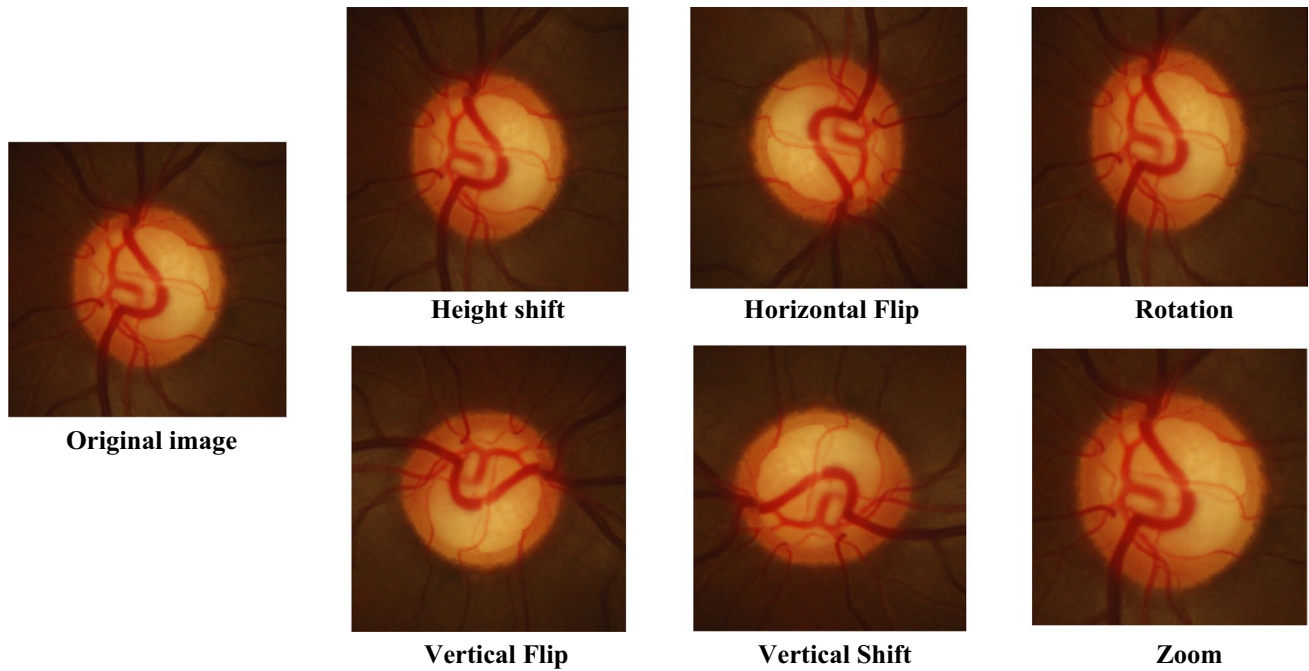


Fig. 11 Augmentation

Gaussian noise is a statistical noise having probability density function (PDF) same as of normal distribution, also known as Gaussian distribution. The PDF of a Gaussian random variable ‘ z ’ in a grey scale image is given in Eq. (11)

$$p_G(z) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(z-\mu)^2}{2\sigma^2}} \quad (11)$$

Here, ‘ z ’ represents the grey level pixel value, ‘ μ ’ is mean value and ‘ σ ’ is the standard deviation.

Further, the salt and pepper noise is an impulsive noise with PDF given in Eq. (12).

$$p_{S\&P}(z) = \begin{cases} P_a & \text{for } z = a \\ P_b & \text{for } z = b \\ 0 & \text{otherwise} \end{cases} \quad (12)$$

In Eq. (12) P_a and P_b represents the PDF of grey scale image for $a=0$ and $b=255$; if $b > a$, grey level $b=255$ will appear as white dot and $a=0$ will appear as black. If either P_a or P_b is zero, the impulsive noise is unipolar. If neither of P_a or P_b is zero and are approximately equal, impulsive noise is salt and pepper. Now, the PDF for image corrupted with both Gaussian and salt and pepper noise is given in Eq. (13)

$$p(z) = p_G(z) + p_{S\&P}(z) \quad (13)$$

Finally, the denoising of image having PDF = ‘ $p(z)$ ’, is performed using adaptive median filter comprising of two levels as given in algorithm below:

Level A:

A1= $z_{\text{median}}-z_{\text{min}}$

A2= $z_{\text{median}}-z_{\text{max}}$

If (**A1**> 0) and (**A2**<0), goto level B

Else increase window size

If (window size<= S_{max}) repeat level A

Else return z_{xy}

Level B:

B1= $z_{xy}-z_{\text{min}}$

B2= $z_{xy}-z_{\text{max}}$

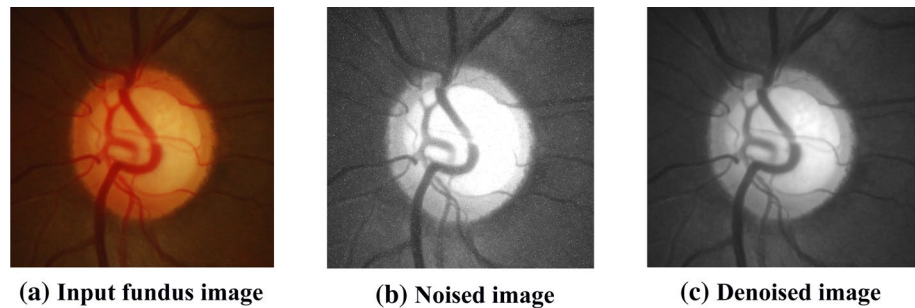
If (**B1**>0) and (**B2**<0), return z_{xy}

Else return z_{median}

Here, z_{min} , z_{max} , z_{median} represents minimum, maximum, median value of grey levels in window. While, z_{xy} is value of grey level and S_{max} is the maximum allowed size in window.

Figure 12 shows the input retinal fundus images followed by greyscale of noised image contaminated with Gaussian, salt and pepper; and greyscale of denoised retinal image filtered using adaptive median filter.

Fig. 12 Denoising of fundus image



4.2 Classification using GC-NET

The input fundus images to the proposed CNN model were cropped to the size 512×512 pixels and further resized to 224×224 pixels in RGB format. Since the original size of the images were quite large, CNN was not able to train due to the high-power computation, therefore, the images were resized to 224×224 pixels, pertaining the optic disc. The proposed model included 76 layers with Input (1) RGB image of size 224×224 pixels, Convolutional (21), Maxpooling (10), Batch Normalization (21), Activation (11), Add (9), Dropout (1), and Dense layers (2). All the layers of convolution involved filters of size (1,1) and (3,3) with 64, 128, 256, 784 and 1024 number of filters in different layers activated using ReLU activation function. The loss of data between layers was reduced by the Add layers in each block to merge the output of the previous block and upcoming block. The maxpool layers were used to reduce the spatial dimensions of the image by selecting the maximum values from the processed image. The batch normalized the previous layer activations by applying a transformation to maintain the mean activation close to 0 and standard deviation close to 1. The activation layers were used to bring non-linearity in the output of previous layers. The linear activation function was used to bring out the linearity in the output of the layers, and the dropout layer was used to prevent the model from overfitting. The proposed model used a dropout rate of 0.2 with global maxpooling layer to convert the 2D output of the previous layer to 1D prediction output. Two dense layers were used to form a fully connected neural network with the unit argument of 32 and 1 with ReLU and sigmoid activation arguments, respectively.

4.3 Designing and Training

The proposed CNN architecture was used for classification between glaucomatous and normal image. Layers were organized in the order, with two convolutional, two batch normalization and an activation layer included in each block with maxpooling layers added in each block to reduce the dimensions of the images. The Add layers were included in each block, excluding block one and the final block to reduce

data loss. A dropout layer was included in each block to prevent overfitting. Finally, a global maxpooling and a dense layer were included in the last block to give a single output unit. The optimizer and loss functions such as `ada_delta` with a learning rate of 0.01 and `binary_crossentropy` were used, respectively, and the logistic regression approach was used to classify the glaucomatous and normal images.

4.4 Prediction

The proposed GC-NET model predicted the retinal image as glaucomatous or normal based on the training provided to the different blocks as shown in Fig. 13 and layers as presented in Table 1.

5 Results and discussion

This section includes the experimental setup and dataset used for experimentation, followed by metrics for performance analysis of different deep learning classifiers such as Xception, InceptionV3, ResNet, DenseNet, VGG16 and the proposed GC-NET. All the experimentation has been performed using the same dataset and the same pre-processing to avoid bias in the results.

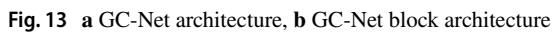
5.1 Experimental setup and datasets used

5.1.1 Experimental setup

The proposed approach for classification of retinal fundus images as normal and abnormal was implemented on python version 3.7 and tensorflow version 1.13.1. Furthermore, the approach was executed using Nvidia GeForce GTX TITAN× GPU and two Intel(R) Xeon(R) E5-2620 2.4 GHz CPUs.

5.1.2 Datasets

- (i) *DRISHTI-GS* It comprises of 101 retinal fundus images with 30 normal images and 71 glaucomatous images acquired using a retinal fundus camera.



(ii) *RIM-One* This dataset comprises of 166 retinal fundus images with labels “glaucoma” or “non-glaucoma” followed by ground truth of cup and disc. The images in the dataset are acquired under different

5.2 Testing Parameters

Experimental evaluation was performed on a total number of 267 images combined from both datasets. This number (267) combined from both datasets was further increased to 1332 for experimental evaluation of deep learning. Training was performed on 60% of images, while remaining 40% of the images were subjected to testing/validation on rest 40%. The same set of pre-processing comprising

of cropping, augmentation and denoising is applied for Xception, InceptionV3, ResNet50, DenseNet121 and VGG16 along with the proposed GC-NET to avoid bias

in performance evaluation. Table 2 presents the model parameters used for training and testing of the above CNN approaches.

Table 1 Descriptions of blocks

Layer (type)	Filter size	Param #	Connected to
First Block			
Input_1 (InputLayer)		0	
conv2d_1 (Conv2D)	3×3	1792	input_1[0][0]
batch_normalization_1		256	conv2d_1[0][0]
activation_1 (Activation)		0	batch_normalization_1[0][0]
Middle Block (repeated 9 times)			
separable_conv2d_1	3×3	8896	activation_1[0][0]
batch_normalization_3		512	separable_conv2d_1[0][0]
activation_2 (Activation)		0	batch_normalization_3[0][0]
conv2d_2 (Conv2D)	3×3	8320	activation_1[0][0]
max_pooling2d_1	3×3	0	activation_2[0][0]
batch_normalization_2		512	conv2d_2[0][0]
add_1 (Add)		0	max_pooling2d_1[0][0] batch_normalization_2[0][0]
Final Block			
separable_conv2d_10	3×3	35,200	add_9[0][0]
batch_normalization_20		512	separable_conv2d_10[0][0]
activation_11 (Activation)		0	batch_normalization_20[0][0]
separable_conv2d_11	3×3	17,664	activation_11[0][0]
batch_normalization_21		512	separable_conv2d_11[0][0]
activation_12 (Activation)0			batch_normalization_21[0][0]
dropout_1 (Dropout)		0	activation_12[0][0]
global_max_pool		0	dropout_1[0][0]
dense_1 (Dense)		4128	global_max_pooling2d_1[0][0]
dense_2 (Dense)		33	dense_1[0][0]

Table 2 Parameters used for training and testing of CNN approaches

Model Parameters/Hyperparameters	Xception	InceptionV3	ResNet50	DenseNet121	VGG16	Proposed approach
Number of layers	135	314	178	430	23	76
Learning rate	1.0	1.0	0.001	1.0	1.0	0.05
Epochs	30	30	30	30	30	30
Batch size	32	32	16	16	16	16
Optimizer	Adadelta	Adadelta	SGD	Adadelta	Adadelta	Adadelta
Activation (hidden layers)	ReLU	ReLU	ReLU	ReLU	ReLU	Linear
Activation (output layer)	Sigmoid	Sigmoid	Sigmoid	Sigmoid	Sigmoid	Sigmoid

Table 3 Confusion matrix

Output of classification	Condition (e.g. Glaucoma/Normal) by ground truth		
	Positive	Negative	Total of Row
Positive	T_p	F_p	$T_p + F_p$
Negative	F_n	T_n	$F_n + T_n$
Total of Column	$T_p + F_n$ (total cases with the given condition)	$F_p + T_n$ (total cases without given condition)	$N = T_p + T_n + F_p + F_n$ (Total cases)

5.3 Performance metrics [44]

The performance of different classifiers was analysed using metrics such as specificity, sensitivity and accuracy. These metrics are derived from the values of True positive (T_p), False positive (F_n), True negative (T_n) and False negative (F_n) derived from the confusion matrix. Thus, from Table 3 of the confusion matrix:

T_p is defined as, prediction as a disease and is actually a disease.

F_p is defined as, prediction as a disease and is actually normal.

T_n is defined as, prediction as a normal and is actually normal.

F_n is defined as, prediction as a normal and is actually as a disease.

Further specificity, sensitivity and accuracy were computed as:

- (i) *Sensitivity* It computes the fraction of actual positives that are correctly recognized as such and is computed using Eq. (14):

$$\text{Sensitivity} = \frac{T_p}{T_p + F_n} \times 100 \quad (14)$$

- (ii) *Specificity* It calculates the fraction of actual negatives which are correctly recognized as such and is computed using Eq. (15):

$$\text{Specificity} = \frac{T_n}{T_n + F_p} \times 100 \quad (15)$$

- (iii) *Accuracy* It is the measure of closeness of result to a correct value and is computed using Eq. (16):

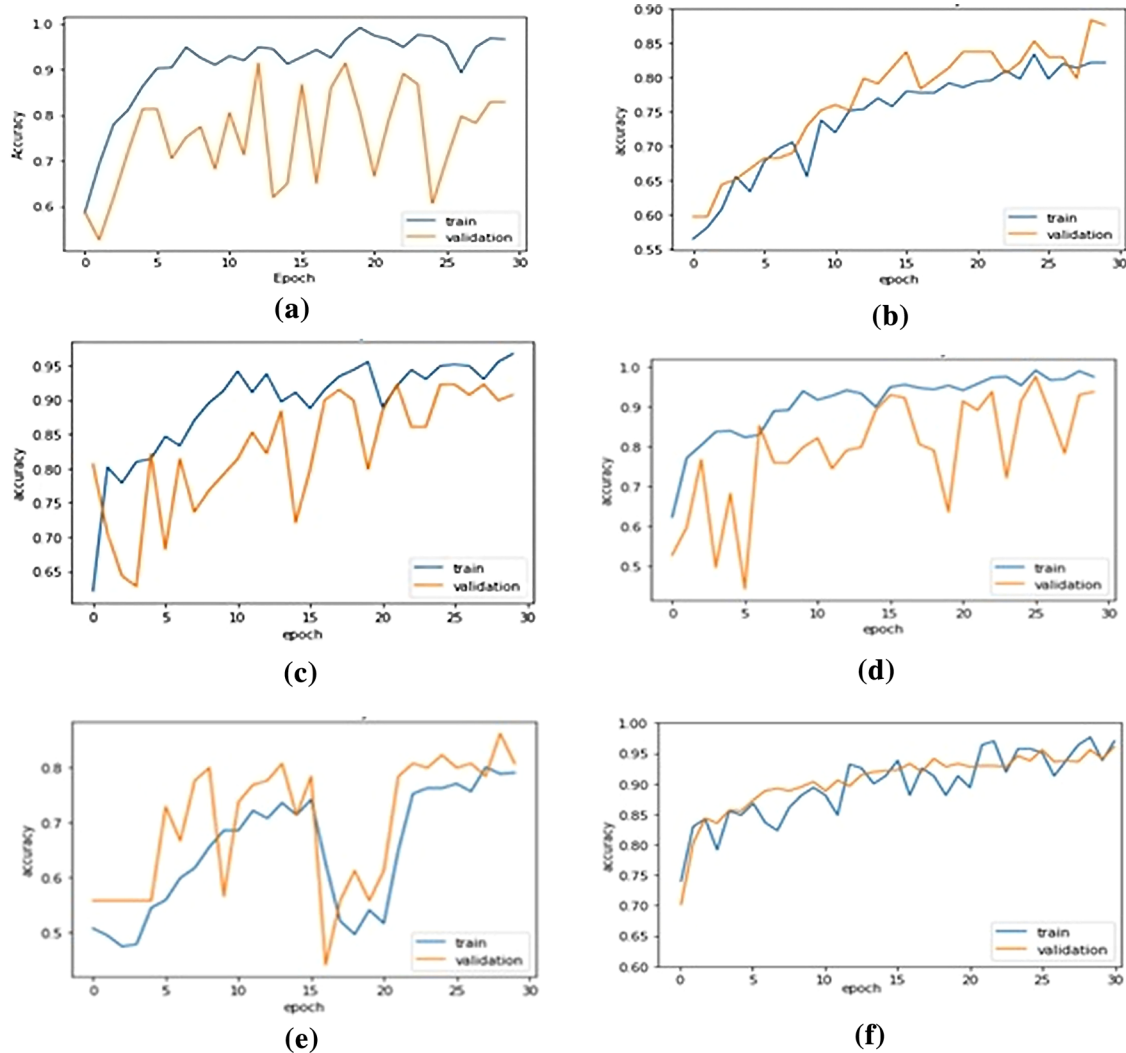


Fig. 14 Training and Validation plots of Convolutional neural networks

$$\text{Accuracy} = \frac{T_p + T_n}{T_p + T_n + F_p + F_n} \times 100 \quad (16)$$

5.4 Performance analysis

The performance of Xception, InceptionV3, ResNet50, DenseNet121, VGG16 and proposed GC-Net architecture was analysed on the Drishti-GS and Rim-One dataset.

Table 4 Performance of deep learning models

S. no.	Architecture	Performance metrics						
		T_p	F_p	T_n	F_n	Accuracy (in%)	Sensitivity (in%)	Specificity (in%)
1.	Xception	0.80	0.07	0.70	0.04	93.17	91.95	90.90
2.	InceptionV3	0.70	0.01	0.76	0.04	90.68	94.59	87.35
3.	ResNet50	0.56	0.01	0.78	0.01	83.22	84.84	82.10
4.	DenseNet121	0.79	0.05	0.72	0.05	93.78	94.04	91.13
5.	VGG16	0.62	0.01	0.72	0.09	83.22	87.32	80.00
6.	Proposed architecture	0.81	0.03	0.76	0.01	97.51	98.78	96.20

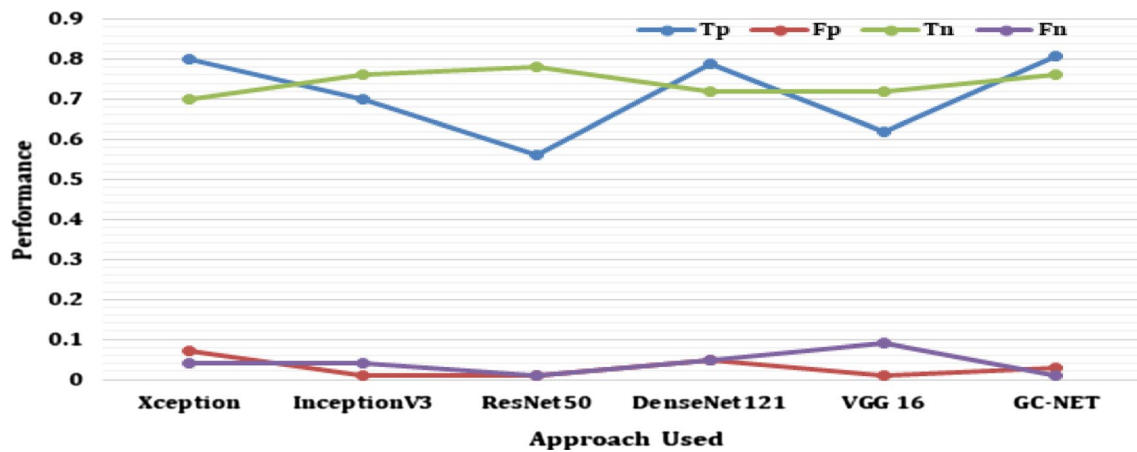


Fig. 15 Plots of TP, Fp, Tn, Fn

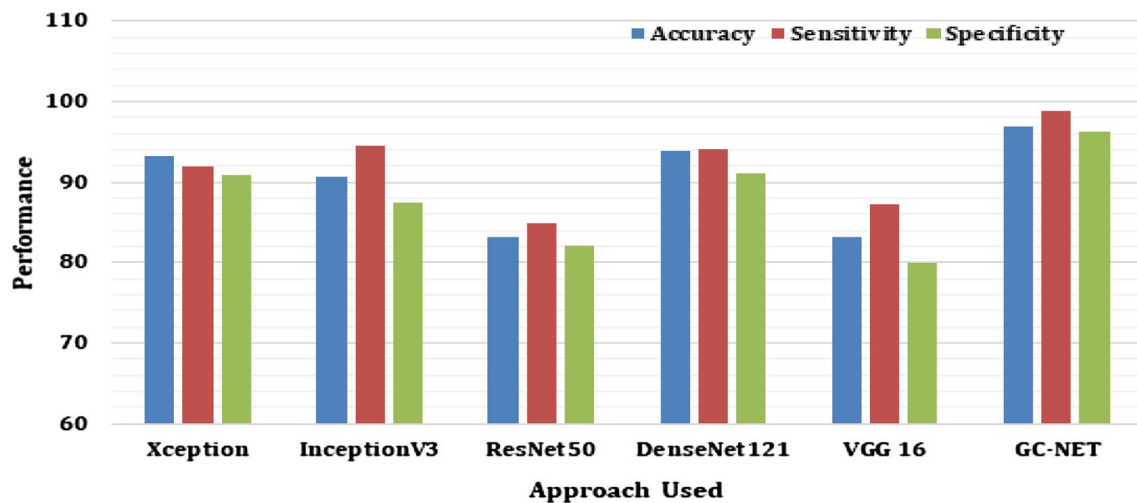


Fig. 16 Plots of accuracy, sensitivity and specificity

Figure 14 illustrates the training and validation plots for each model. Validation means validating the architecture after each epoch on the validation dataset. Here, the time interval between each epoch for the Xception is 12 s, InceptionV3 is 16 s, ResNet50 is 14 s, DenseNet121 is 19 s, VGG16 is 15 s, and the proposed GC-Net is 30 s.

As observed from the plot of Inception V3 (Fig. 14a), there is more gap between validation accuracy and training accuracy. This suggest that the model could be overfitting. Also, the increase in accuracy has not been consistent. While, in the plot for ResNet (Fig. 14b), there are large gaps between training and validation at few epochs. Also, the training accuracy is way lower than validation accuracy at the final epoch. Further, in plots of Xception (Fig. 14c), it can be seen that the validation accuracy is lower than the training accuracy. This suggest that the model overfits and the peaks are not increasing consistently. In DenseNet (Fig. 14d), the training accuracy is increasing consistently but validation accuracy is not following the same trend. Also, there is a reduction in validation accuracy after a few epochs, which is not a desirable. Hence the model underperforms and is not good for prediction of abnormality. In the case of VGG 16 (Fig. 14e), the curves of both training and validation follow a similar path. But after a few epochs, both the accuracies decrease sharply. This is also not a desirable learning trend and is thus not suitable for accurate prediction. Finally, it could be observed that the GC-Net curve shows a steady increase in accuracy for each epoch, which signifies that the model is well trained and performs prediction with improved accuracy. However, the other plots with sudden peaks indicate fall in accuracy with an increasing number of epochs, suggesting predictions with reduced accuracy. The validation curves show the accuracy achieved by the models after each epoch. Further, the experimental results in terms of Tp , Fp , Tn , Fn , sensitivity, specificity and accuracy are presented in Table 4 along with Fig. 15 representing plots of Tp , Fp , Tn and Fn . Whereas, Fig. 16 gives the plots of sensitivity, specificity and accuracy. The execution time (i.e. training time and testing time) for the proposed GC-Net architecture has been observed to be 12.75 min and 4 s, respectively.

The proposed GC-NET model comprises of 76 layers deep neural network and separable convolution which is the key feature that offers improved classification in comparison with state of the art. The separable convolution layer extract features after separating the channels from each block. Sigmoid is here used as an activation function, which is positioned before the last fully connected layer to perform binary classification. Also, a linear activation function is added at the end of each block for optimal classification with a learning rate of 0.15, batch size of 16, ada_delta optimizer and 30 epochs. The algorithm for the model involves the

extraction of features using separable convolutional layers which helps in extracting features across the three channels, i.e. red, green and blue. This prevents data loss, therefore an increase in the accuracy of the model is seen. The proposed approach also involves an increase in the filter size over the state of art approaches. The increase in filter size ensures more feature extraction, hence an increase in the accuracy of prediction. The skip connection after each block helps in further boosting the accuracy of the model since the basic principle behind skip connection is the retainment of data from the previous block and combining it with the output of the current block, which then acts as the input to the next block. All the factors mentioned above resulted in a more accurate and efficient classification between a glaucomatous and an ordinary eye.

Based on the experimental analysis, accuracy of Xception is found to be 93.17% with a sensitivity of 91.95%, specificity of 90.9%, Tp of 0.80, Fp of 0.07, Tn of 0.70 and Fn of 0.04. On the other hand, InceptionV3 has the accuracy of 90.68%, sensitivity of 94.59% and specificity of 87.35% with 0.70 Tp , 0.01 Fp , 0.76 Tn and 0.04 Fn . The accuracy for Resnet50 was observed to be 83.22% with sensitivity of 84.84% and specificity of 82.10% followed by 0.56 Tp , 0.01 Fp , 0.78 Tn and 0.01 Fn . DenseNet121 achieved the accuracy of 93.78%, sensitivity of 94.04% and specificity of 91.13% with 0.79 Tp , 0.05 Fp , 0.72 Tn and 0.05 Fn . VGG16 got the accuracy of 83.22%, sensitivity of 87.32% and a specificity of 80% with 0.62 Tp , 0.01 Fp , 0.72 Tn and 0.09 Fn . Finally, the GC-Net architecture proposed in this study achieved the accuracy of 97.51%, sensitivity of 98.78% and specificity of 96.20% with 0.81 Tp , 0.03 Fp , 0.76 Tn and 0.01 Fn . Thus, it can be stated that the GC-NET outperforms Xception, InceptionV3, ResNet50, DenseNet121, VGG16 with highest values of sensitivity, specificity and accuracy along with optimal values of Tp , Fp , Tn , Fn .

6 Conclusion

Glaucoma as a retinal disorder has adversely affected loss of vision for a large segment of human population across the globe and is expected to affect a greater number of people if left untreated. Early diagnosis is a key aspect for prevention of this vision deterioration, which if not treated timely could further escalate towards permanent vision loss. Due to the manual and slower approach of traditional diagnosis methods, early detection and diagnosis of this disease gets severely delayed. Also, the accuracy of state of the art is not up to the mark and hence not clinically acceptable. The proposed GC-NET model extracts features based on CNN and self learns to classify the retinal fundus image samples into normal and abnormal cases. It performs better than state of the art with an accuracy of 97.51%, sensitivity of 98.78%

and specificity of 96.20% with 0.81 T_p , 0.03 F_p , 0.76 T_n and 0.01 F_n which can be clinically acceptable. Hence, the approach can be tested on a large-scale clinical image to make it globally available for use in real scenarios, so that it can be used as a second opinion for diagnosis of glaucoma by the expert ophthalmologists.

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